

HIDDEN BARGAINS

Finding undervalued players

Prepared by:

Afonso Marques	David Afonso	Guilherme Valido	Martim Coelho	Miguel Marto
20241691	20241692	20241755	20241797	20241798

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EXECUTIVE SUMMARY

This report establishes a data-driven framework to identify undervalued forwards across Major and Secondary leagues. Integrating performance metrics from FBref and valuations from Transfermarkt, we applied Principal Component Factoring (PCF) and Cluster Analysis to categorize players into distinct tactical profiles based on three extracted dimensions: Playmaking & Workrate, Pure Finishing Threat, and Error Proneness. The analysis revealed structural differences between ecosystems, identifying "Complete Finishers" as an elite commodity in Major Leagues, whereas Secondary Leagues are dominated by mediocrity or dominant "Alien" profiles. By benchmarking market values against cluster-specific medians, we successfully identified multiple "hidden bargains", elite performers priced significantly below their peer group, such as Franck Honorat and Mohamed Nassoh. The study concludes that combining statistical profiling with financial analysis offers clubs a scalable, evidence-based strategy to exploit market inefficiencies and mitigate recruitment risk.

INTRODUCTION

The modern football transfer market has become a highly competitive and financially complex landscape in which player valuation often diverges from actual on-field performance. This discrepancy creates opportunities for clubs to identify **“hidden bargains”**, forwards who deliver strong statistical output despite possessing comparatively low market value. So, **which forwards outperform their peers yet remain undervalued relative to others with similar performance profiles?**

To address this question, we adopt a rigorous, data-driven analytical framework designed to bridge the gap between football performance and economic valuation. The study focuses on forwards who played at least 35% of game time during the 2024/2025 season, ensuring statistical reliability and reducing noise from small sample sizes. Players are divided into two broad competitive contexts:

1. **Major European Leagues** (Premier League, Bundesliga, Serie A, La Liga, Ligue 1).
2. **Secondary Leagues** (Liga Portugal, Eredivisie, Jupiler Pro League, Brasileirão, and the EFL Championship).

This separation allows us to account for differences in league competitiveness and potential inflation of statistics in less competitive environments.

INTRODUCTION

This methodological segmentation is crucial to mitigate contextual bias, accounting for the disparity in league competitiveness and the potential inflation of performance metrics in lower-ranked competitions. The classification of these competitive clusters was empirically grounded in the Opta Power Rankings, as illustrated in Figure 1:



Fig. 1 - League competitiveness comparison based on Opta Power Rankings.

INTRODUCTION

Performance data was collected from **FBRef**, incorporating advanced metrics such as Expected Goals (xG), Shot-Creating Actions (SCA), Progressive Passes, and Defensive Work Rate. **Transfermarkt** was used for market valuation. After merging the data in **SAS**, we applied **Principal Component Factoring** (PCF) analysis to reduce the numerous performance variables into composite performance scores that summarize each player's overall footballing contribution.

Clustering was then conducted using only these performance scores, allowing players to be grouped based on similarity in on-field output. Finally, within the clusters representing the highest-performing forwards, we compared each player's market value to the cluster average to identify undervalued individuals.

Through this structured approach, the report aims to offer an objective, reproducible method for identifying **hidden bargains** across leagues of differing competitive levels, providing practical **insights for scouting departments and data recruitment teams**.

METHODOLOGY

DATA COLLECTION

We collected player-level performance statistics from **FBref**. To ensure statistical reliability and avoid noise from small sample sizes, we restricted the dataset to players who logged **at least 35% of the total avaliable minutes** in the 2024/2025 season.

Variables	Meaning
Gls	Goals per 90
Sh	Shots per 90
G/Sh	Goals by shot
G - xG	Goals minus Expected goals per 90
Ast	Assists per 90
PPA	Passes to penalty area per 90
CrsPA	Crosses into the penalty area per 90
SCA	Shot creating actions per 90
GCA	Goal creating actions per 90
Att 3rd	Tackles in the attacking third per 90
Tkl + Int	Tackles + Interceptions
AttPen	Touches in the penalty area
TOSucc	Take-ons succeded per 90
PrgC	Progressive carries per 90
Mis	Number of times the player failed to control a ball per 90
Dis	Number of times a player lost the ball per 90
Aewon	Aerials won per 90

Fig. 2 - Metadata

METHODOLOGY

DATA PREPARATION AND INTEGRATION

We used Excel, Python and SAS for this project. Because the raw datasets came from multiple leagues and contained slight structural differences, we used SAS to merge and clean all observations. This included:

- Aligning variable names and formats
- Removing duplicates
- Ensuring consistent player identifiers
- Filtering players by position and minutes played

- **Standardizing the data:**

Since the selected variables operate on vastly different scales (e.g., Chances created per 90 vs. Goals per 90), standardization was applied to ensure that no single variable would disproportionately influence the variance-based calculations in PCF and Clustering.

- **Missing Values:**

There were no missing values to treat since the database from FBRef had all the data needed for every observation in our dataset.

The final result was a single, unified dataset containing forward performance metrics. The market values are going to be taken into account after the cluster formation.

Player	Gls	Sh	G/Sh	G-xG	Ast	PPA	CrsPA	SCA	GCA	Att 3rd	Tkl+Int	Att Pen	TOSucc	PrgC	Mis	Dis	AeWon
Alexander Isak	0.75	3.10	0.20	0.09	0.20	1.01	0.03	3.01	0.46	0.23	0.49	6.18	1.37	2.71	2.16	1.80	0.85
Antoine Semenyo	0.31	3.51	0.09	0.03	0.14	1.26	0.22	3.71	0.37	0.37	1.85	5.37	1.94	3.93	2.36	2.16	1.85
Beto	0.47	2.88	0.16	-0.01	0.00	0.47	0.06	1.53	0.06	0.24	1.06	5.06	1.12	0.88	4.18	2.18	5.82
Brennan Johnson	0.45	2.23	0.20	0.02	0.12	1.45	0.25	1.90	0.29	0.41	1.78	5.17	0.79	2.48	1.07	0.70	0.29
Bryan Mbeumo	0.53	2.08	0.19	0.20	0.18	1.56	0.50	3.80	0.53	0.42	1.66	4.43	1.37	3.43	2.80	1.24	0.90
Cameron Archer	0.13	1.75	0.07	-0.23	0.00	0.44	0.00	1.25	0.00	0.13	0.94	3.25	0.94	1.38	2.50	1.63	0.56
Chris Wood	0.61	1.98	0.26	0.20	0.09	0.36	0.03	1.34	0.15	0.06	0.33	3.19	0.12	0.43	2.28	0.85	2.86

Fig. 3 - Sample of the dataset

METHODOLOGY

DATA PREPARATION AND INTEGRATION

- **Summary Statistics:**

An analysis of the summary statistics (Fig. 4) reveals significant variance in metrics such as Shot-Creating Actions (SCA), Progressive Carries (PrgC), and Aerial Duels Won (AeWon). These high-variance variables proved to be crucial differentiators in our subsequent analysis, effectively distinguishing between player archetypes such as "Playmakers" (high SCA/PrgC and negative AeWon) and "Finishing Threats" (low PrgC/SCA and positive AeWon).

	count	mean	std	min	25%	50%	75%	max	variance	median
Gls	138.0	0.420435	0.234053	0.00	0.2800	0.365	0.5300	1.15	0.054781	0.365
Sh	138.0	2.559928	0.817483	1.08	2.0425	2.415	3.0550	5.00	0.668278	2.415
G/Sh	138.0	0.144348	0.059511	0.00	0.1000	0.140	0.1800	0.30	0.003542	0.140
G-xG	138.0	0.015362	0.118119	-0.31	-0.0600	0.020	0.0875	0.44	0.013952	0.020
Ast	138.0	0.150290	0.112742	0.00	0.0700	0.120	0.2000	0.56	0.012711	0.120
PPA	138.0	0.813623	0.651065	0.00	0.3425	0.530	1.1775	3.00	0.423886	0.530
CrsPA	138.0	0.156087	0.209854	0.00	0.0000	0.070	0.1975	1.06	0.044039	0.070
SCA	138.0	2.624203	1.144531	0.50	1.7900	2.350	3.2875	5.93	1.309950	2.350
GCA	138.0	0.345072	0.203406	0.00	0.2100	0.300	0.4600	1.30	0.041374	0.300
Att 3rd	138.0	0.233551	0.136600	0.00	0.1400	0.200	0.3100	0.66	0.018660	0.200
Tkl+Int	138.0	1.064493	0.636718	0.16	0.6525	0.925	1.3975	4.13	0.405410	0.925
Att Pen	138.0	4.452464	1.851374	0.80	3.3225	4.420	5.7250	9.97	3.427585	4.420
TOSucc	138.0	1.017246	0.780865	0.07	0.4500	0.765	1.3675	5.08	0.609750	0.765
PrgC	138.0	2.171449	1.660108	0.14	0.9200	1.460	3.0425	6.81	2.755957	1.460
Mis	138.0	2.549493	0.721967	1.06	2.1150	2.525	2.9900	4.73	0.521237	2.525
Dis	138.0	1.419928	0.478914	0.36	1.1250	1.390	1.6600	2.96	0.229358	1.390
AeWon	138.0	1.771159	1.585877	0.03	0.5825	1.325	2.3575	9.36	2.515004	1.325

Fig. 4 - Summary Statistics

METHODOLOGY

DATA PREPARATION AND INTEGRATION

- **Outliers:**

We chose to identify outliers **only after performing the dimensionality reduction** because the factor scores provide a more meaningful and consolidated view of player performance than any single raw variable. Individual metrics, such as shots, expected goals, passes, or pressures capture only one aspect of behaviour, and a player might appear extreme in one variable but perfectly normal when considering the full multivariate structure. By reducing the data to a small number of principal components, each summarising a broader performance dimension, we were able to evaluate players based on their **overall profiles** rather than isolated statistics.

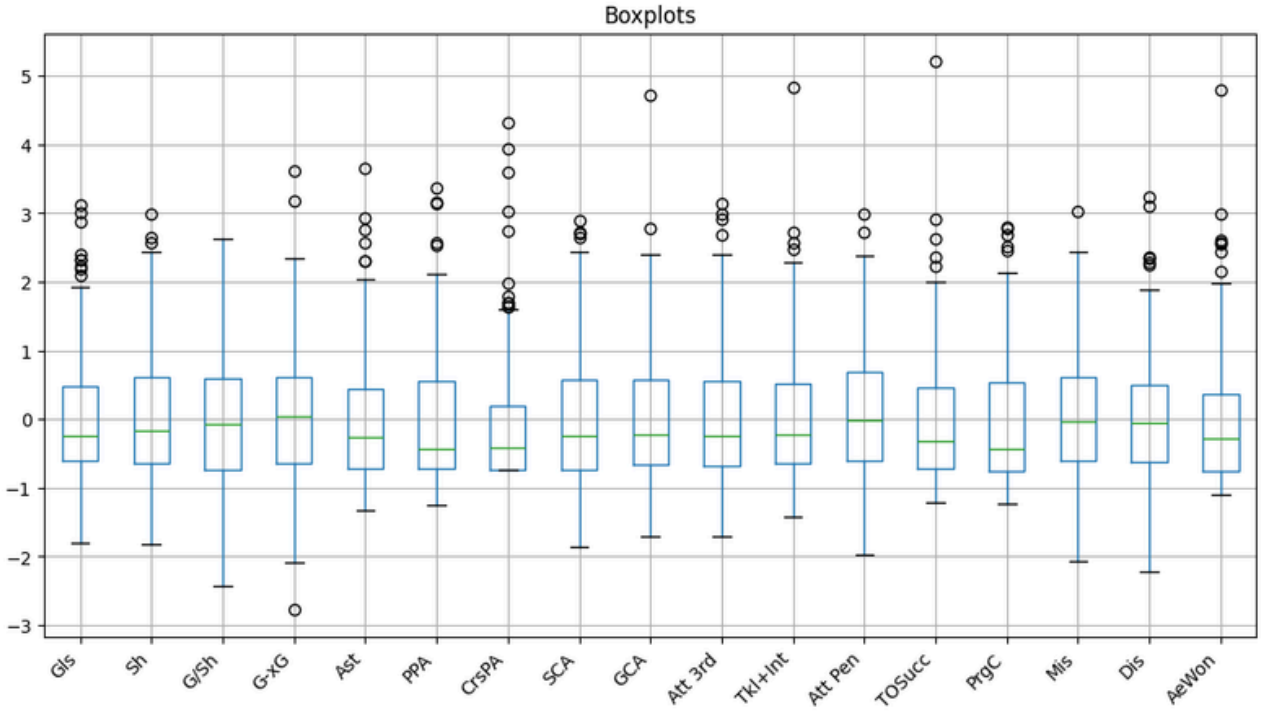


Fig. 5 - Box Plot

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METHODOLOGY

VARIABLE SELECTION

The performance variables, given in the image provided, capture multiple aspects of forward play, such as:

- Scoring efficiency
- Shot volume and quality
- Chance creation and expected assists
- Possession involvement
- Passing quality in the attacking third
- Ball progression
- Defensive contribution from the front

These variables serve as the basis for evaluating **overall performance**, but they are often correlated. For example, players who take more shots often generate higher expected goals, and those who progress the ball frequently may also create more chances. This correlation, in addition of motivating the need for **dimensionality reduction**, it raised the problem of **multicollinearity**.

METHODOLOGY

VARIABLE SELECTION

- **Correlation Matrix:**

In the correlation matrix (Fig. 6), we observed consistent and meaningful correlations among performance variables. For instance, the strongest correlation was observed between SCA and Passes into Penalty Area (PPA) (r = 0.84), indicating a strong shared variance between creative passing and shot generation. These patterns confirm the presence of underlying performance dimensions suitable for factor extraction.

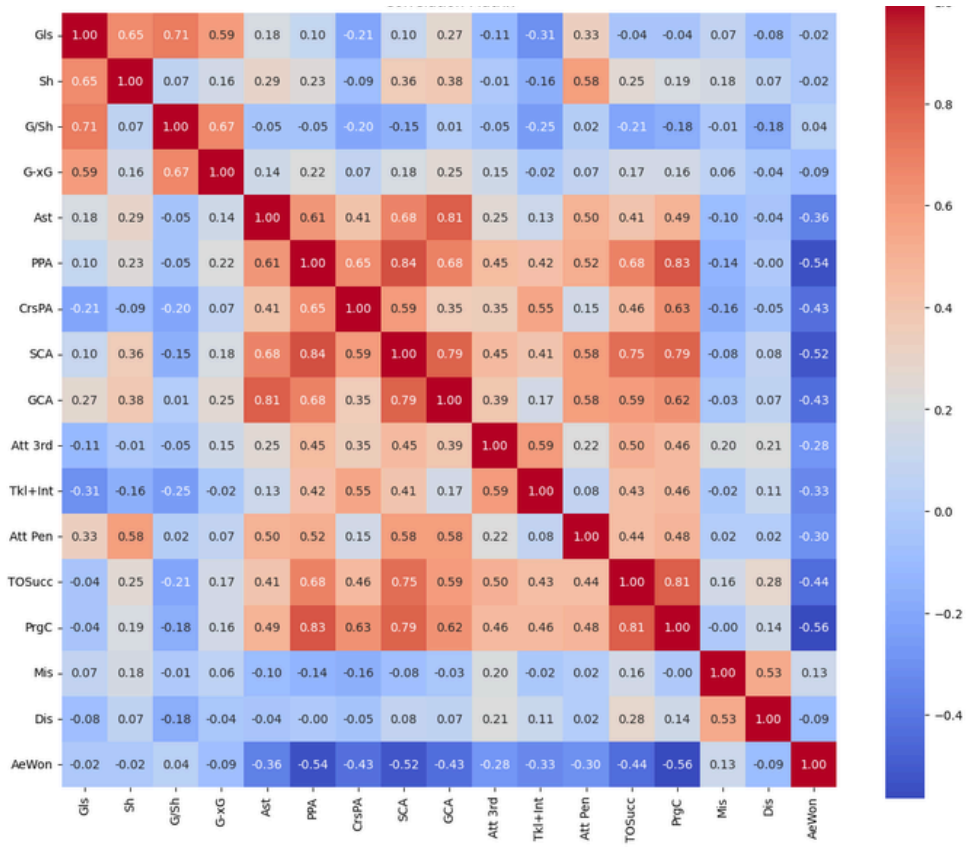


Fig. 6 - Correlation Matrix

METHODOLOGY

VARIABLE SELECTION

In the correlation matrix, we observed that no pair of variables was correlated strongly enough to raise concerns about multicollinearity. However, the variables did display **consistent and meaningful** correlations with one another, indicating shared variance across several dimensions of performance. This pattern suggests that **Principal Component Factor (PCF) analysis** is the most appropriate method for dimensionality reduction. Unlike PCA, which focuses solely on maximizing total variance, PCF extracts factors based on the common variance shared among variables, making it more suitable when the goal is to **uncover underlying performance constructs**. Because our objective is to summarize players' statistical profiles into interpretable performance dimensions, leveraging the structure of these correlations through **PCF allows us to retain the meaningful relationships between variables while improving interpretability.**

METHODOLOGY

DUAL DATASET ANALYSIS

Given the methodological consistency applied to both datasets (Major Leagues and Secondary Leagues), preliminary validation outputs, including correlation matrices, KMO tests, Scree Plots, and hierarchical clustering diagnostics (Dendrograms and R-squared plots), yielded structurally similar patterns across both groups.

Therefore, to ensure conciseness and focus on actionable insights, detailed outputs for these preliminary steps will be presented primarily for the Major Leagues dataset as a representative example. However, distinct results emerged during the Cluster Interpretation phase. Consequently, this section will be explicitly detailed for both groups, highlighting the critical divergences in player profiles and market valuations that drive the study's conclusions.

METHODOLOGY

DIMENSIONALITY REDUCTION THROUGH PCF

Firstly we analyzed the **KMO** of the variables of the dataset to make sure they are suitable for the reduction.

And with an overall KMO of **0,81** and none of the variables having a value lower than 0.50, we could start the process.

Using **Kaiser's criterion**, we initially identified **four factors** with eigenvalues greater than 1, suggesting that a four-factor solution could adequately represent the shared structure of the dataset.

```
KMO Overall: 0.81

Individual KMO:
Gls         0.54
Sh          0.52
G/Sh        0.52
G-xG        0.79
Ast         0.84
PPA         0.92
CrsPA       0.87
SCA         0.92
GCA         0.85
Att 3rd     0.80
Tkl+Int     0.82
Att Pen     0.89
TOSucc      0.89
PrgC        0.89
Mis         0.55
Dis         0.64
AeWon       0.90
```

Fig. 7 - KMO

The **Pearson Criterion** indicated that a three-factor solution already captured approximately 64% of the total common variance, with additional factors offering diminishing returns.

To further refine the decision, we examined the **Scree Plot**, which displayed a clear inflection after the **third or fourth factor**. This visual pattern suggested that both three and four factors were reasonable solutions.

METHODOLOGY

DIMENSIONALITY REDUCTION THROUGH PCF

```
Eigenvalues:
 [6.30026306 2.91442416 1.79446712 1.52661072 0.77081699 0.69953177
 0.6181248  0.48479115 0.42726227 0.3531401  0.27591681 0.22580784
 0.17500045 0.1478054  0.13838813 0.09340218 0.05424706]
```

Fig. 8 - Eigenvalues. Kaiser Criterion

```
Cumulative variance:
[0.37060371 0.54204042 0.64759731 0.73739794 0.78274012 0.82388905
 0.86024933 0.88876646 0.91389953 0.93467248 0.95090288 0.96418569
 0.97447984 0.98317427 0.99131475 0.996809   1.        ]
```

Fig. 9 - Cumulative Variance. Pearson Criterion

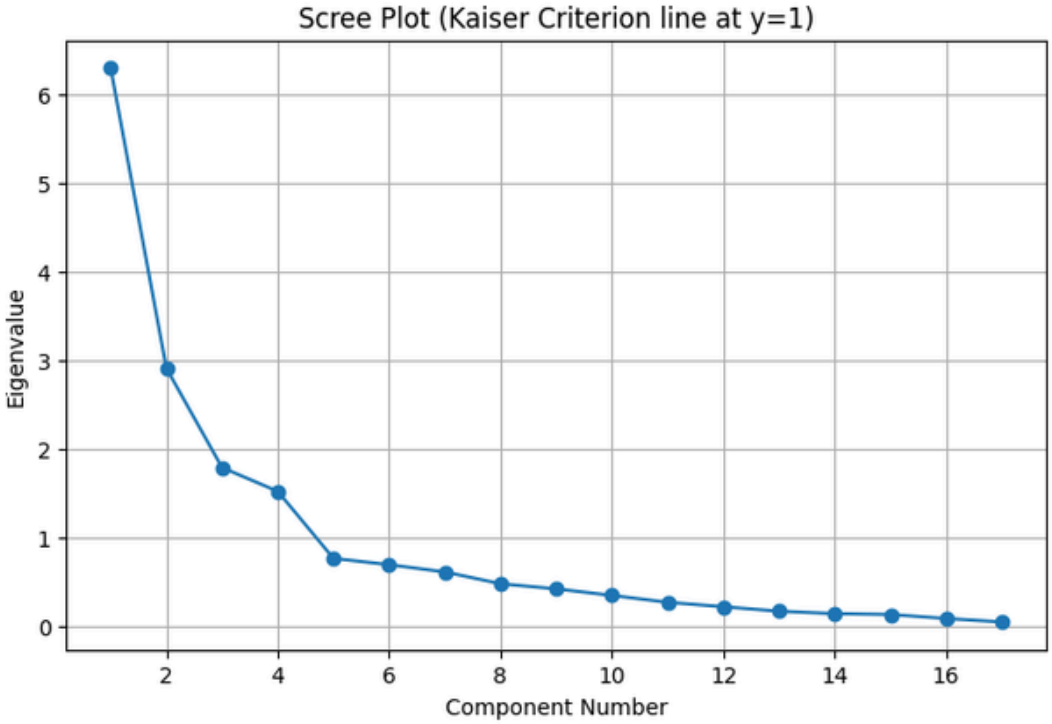


Fig. 10 - Scree Plot

METHODOLOGY

DIMENSIONALITY REDUCTION THROUGH PCF

After determining that both three and four factor solutions were statistically viable, we examined the factor loadings to evaluate the interpretability and added informational value of each structure. We first inspected the **loading matrix** for the **three-component PCF solution**, followed by the matrix obtained when extracting four components. The comparison showed that the first three factors **consistently captured meaningful and interpretable** performance dimensions, each grouping together coherent sets of offensive, creative, and ball-progression metrics.

However, when introducing a fourth factor, we observed that this component contributed **very little** unique information. Its strongest loading, occurred only on the variable **G-xG**, while all remaining variables loaded more strongly onto one of the existing three components.

Because this fourth factor did not represent a distinct or interpretable performance dimension, and added minimal explanatory value, we concluded that retaining three principal components **offered a more stable and meaningful** factor structure for summarizing player performance.

METHODOLOGY

DIMENSIONALITY REDUCTION THROUGH PCF

Image of both loading matrices:

	F1	F2	F3
Gls	0.111013	0.935185	0.034345
Sh	0.333771	0.614339	0.284233
G/Sh	-0.114367	0.736473	-0.152771
G-xG	0.226964	0.606980	-0.044585
Ast	0.711406	0.189860	-0.178724
PPA	0.901420	-0.004298	-0.169047
CrsPA	0.657035	-0.357155	-0.281499
SCA	0.931680	0.019954	-0.046908
GCA	0.820993	0.258123	-0.026144
Att 3rd	0.570796	-0.230990	0.260214
Tkl+Int	0.500144	-0.516928	0.012726
Att Pen	0.627845	0.352805	0.083084
TOSucc	0.816794	-0.129210	0.257743
PrgC	0.885030	-0.145234	0.014329
Mis	-0.017980	0.060364	0.857913
Dis	0.129161	-0.152276	0.802217
AeWon	-0.618026	0.106354	0.153449

Fig. 11 - Three Factor' Loadings

	F1	F2	F3	F4
Gls	0.111013	0.935185	0.034345	0.094640
Sh	0.333771	0.614339	0.284233	-0.434325
G/Sh	-0.114367	0.736473	-0.152771	0.550088
G-xG	0.226964	0.606980	-0.044585	0.630695
Ast	0.711406	0.189860	-0.178724	-0.256967
PPA	0.901420	-0.004298	-0.169047	0.034397
CrsPA	0.657035	-0.357155	-0.281499	0.192699
SCA	0.931680	0.019954	-0.046908	-0.104786
GCA	0.820993	0.258123	-0.026144	-0.162542
Att 3rd	0.570796	-0.230990	0.260214	0.402155
Tkl+Int	0.500144	-0.516928	0.012726	0.367067
Att Pen	0.627845	0.352805	0.083084	-0.393329
TOSucc	0.816794	-0.129210	0.257743	0.038226
PrgC	0.885030	-0.145234	0.014329	0.046272
Mis	-0.017980	0.060364	0.857913	0.125384
Dis	0.129161	-0.152276	0.802217	0.073202
AeWon	-0.618026	0.106354	0.153449	-0.105063

Fig. 12 - Four Factor' Loadings

METHODOLOGY

DIMENSIONALITY REDUCTION THROUGH PCF

After selecting the three-factor solution, we applied a **Varimax rotation** to enhance interpretability by maximizing the separation of high and low loadings across factors. The rotated structure provided **clearer and more distinct** performance dimensions, allowing each factor to represent a coherent aspect of forward play.

Based on the rotated loadings, we interpreted the components in the following way:

- **F1 – Playmaking & Work Rate (Winger Profile):**
  This factor is characterized by high loadings on creative and possession-involvement metrics such as PPA, SCA, GCA, progressive carries, touches in the attacking third, and defensive actions. Players scoring high on F1 typically align with a winger-style profile, contributing heavily to build-up, ball progression, and chance creation while also engaging in off-ball defensive work.

- **F2 – Pure Finishing Threat (Striker Profile):**
  This factor is dominated by strong loadings on goals, shots, shot accuracy (G/Sh), expected goals (xG), and touches in the penalty area. High scorers on F2 match the archetype of a traditional striker, whose primary focus is generating and converting goalscoring opportunities.

- **F3 – Error Proneness:**
  This factor loads heavily on miscontrols and dispossessions, representing players who are more likely to lose the ball or commit technical mistakes during possession.

METHODOLOGY

DIMENSIONALITY REDUCTION THROUGH PCF

This rotated factor structure provided a clear, interpretable foundation for constructing player performance scores and subsequently clustering players according to their underlying performance profiles.

	F1	F2	F3
Gls	-0.015205	0.942255	-0.000305
Sh	0.222430	0.663094	0.283610
G/Sh	-0.193935	0.708711	-0.197302
G-xG	0.149407	0.629935	-0.052770
Ast	0.694513	0.276903	-0.122732
PPA	0.906336	0.110371	-0.086753
CrsPA	0.721736	-0.275746	-0.203857
SCA	0.921375	0.142741	0.036323
GCA	0.779214	0.364545	0.035681
Att 3rd	0.568457	-0.143247	0.321269
Tkl+Int	0.559530	-0.444605	0.082301
Att Pen	0.565855	0.436328	0.122438
TOSucc	0.798249	-0.009625	0.336133
PrgC	0.890940	-0.024942	0.100894
Mis	-0.107591	0.087602	0.848960
Dis	0.070569	-0.105273	0.816921
AeWon	-0.638419	0.028064	0.091914

Fig. 13 - Three Factor' Loadings with Varimax rotation

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METHODOLOGY

DIMENSIONALITY REDUCTION THROUGH PCF

The model achieved a Root Mean Square Residual (RMSR) of 0.087, falling within the acceptable threshold (< 0.10) and indicating a good fit. Following this validation, factor scores were computed for each player, quantifying their performance in each of the three dimensions. These scores serve as the input for the subsequent cluster analysis. The histograms below illustrate the distribution of these scores:

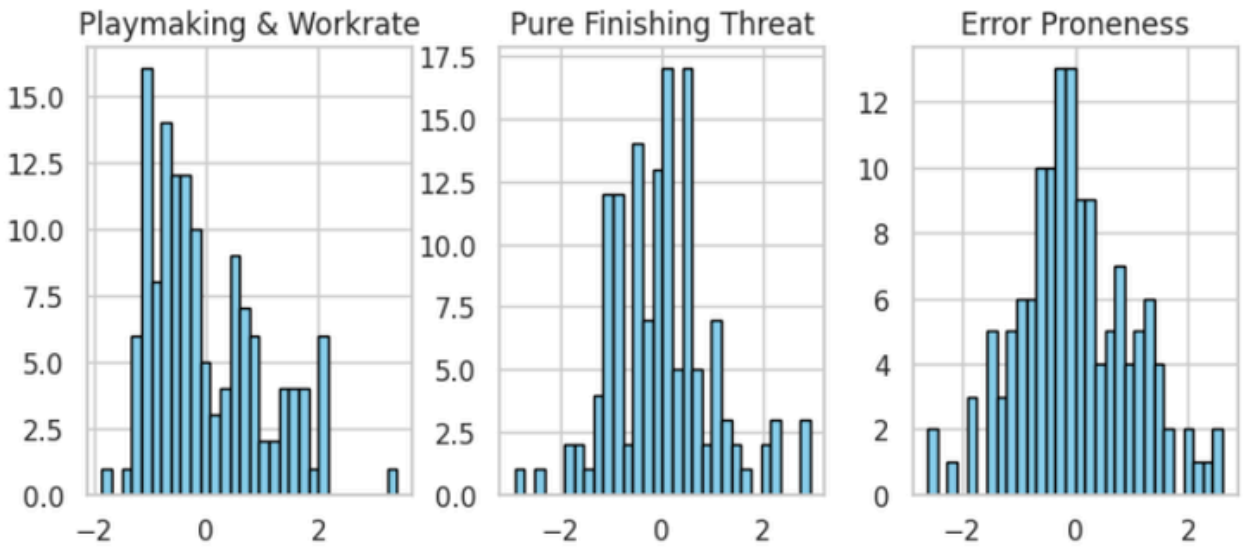


Fig. 14 - Histograms of the three factors

All three histograms show **roughly normal** distributions, meaning the factor scores are well-balanced across players. The only small observation is that **Pure Finishing Threat** is slightly more concentrated around the mean, suggesting fewer extreme finishers in the dataset. Overall, the distributions support the **stability and interpretability** of the components.

METHODOLOGY

OUTLIER ANALYSIS: IDENTIFYING THE “ALIENS”

Following the Factor Analysis, we scrutinized the distribution of factor scores to identify observations that deviated substantially from the general population. This step revealed a group of multidimensional outliers, players whose performance metrics were so extreme that they could not be accurately categorized into standard clusters.

Major European Leagues: In the Big 5 dataset, we identified seven distinct outliers: Jesper Lindstrøm, Franck Honorat, Omar Marmoush, Patrik Schick, M’Bala Nzola, Ousmane Dembélé, and Lamine Yamal. A critical distinction was made here between negative outliers (players with extreme scores in Error Proneness or consistently low output) and positive outliers, whom we designated as "Aliens". The detection of Ousmane Dembélé (24/25 Ballon d'Or winner) and Lamine Yamal (Runner-up) as statistical anomalies serves as a powerful validation of our Factor Analysis model. It confirms that our extracted factors correctly identified the absolute elite of world football as being "off the charts" compared to the average player.

Player	Playmaking & Workrate	Pure Finishing Threat	Error Proneness
Omar Marmoush	2.128932	2.925051	2.475442
Ousmane Dembélé	2.065622	2.811699	0.133838
Patrik Schick	-1.079333	2.796960	-0.044956
Lamine Yamal	3.393932	0.206574	1.621513
M'Bala Nzola	-0.548236	-0.156836	2.612847
Franck Honorat	1.793703	-0.851557	-2.611771
Jesper Lindstrøm	0.617044	-2.899714	-0.090516

👽 Aliens (5): Franck Honorat, Omar Marmoush, Patrik Schick, Ousmane Dembélé, Lamine Yamal
🔻 Negative Outliers (2): Jesper Lindstrøm, M'Bala Nzola

Fig. 15 - Outliers of the Big5

METHODOLOGY

OUTLIER ANALYSIS: IDENTIFYING THE “ALIENS”

Secondary Leagues: A similar pattern emerged in the secondary leagues, where six players (Tolu Arokodare, Estêvão Willian, Ryan Hedges, Noa Lang, Wout Weghorst, and Viktor Gyökeres) detached significantly from the sample. Here, too, we distinguished between statistical underperformers (e.g., Ryan Hedges, due to extremely low finishing scores) and the elite "Aliens" dominating their respective leagues. A standout case is Viktor Gyökeres: with a Pure Finishing Threat score of 4.5 (more than four standard deviations above the mean), combined with solid playmaking (0.55) and exceptionally low error proneness (-1.0), he represents the absolute statistical ceiling of our dataset, a true anomaly that dominated the 2024/2025 Portuguese League.

Player	Playmaking & Workrate	Pure Finishing Threat	Error Proneness
Viktor Gyökeres	0.554912	4.504145	-1.108324
Tolu Arokodare	-0.932076	2.759065	1.141685
Wout Weghorst	-0.616704	2.711414	-1.550719
Noa Lang	3.132365	2.686757	-1.490330
Estêvão Willian	2.220318	2.524382	0.769495
Ryan Hedges	1.076364	-2.583520	0.488313

```
------------------------------
👽 Aliens (5): Tolu Arokodare, Estêvão Willian, Noa Lang, Wout Weghorst, Viktor Gyökeres
🔻 Negative Outliers (1): Ryan Hedges
------------------------------
```

Fig. 16 - Outliers of the Secondary Leagues

METHODOLOGY

OUTLIER ANALYSIS: IDENTIFYING THE “ALIENS”

Methodological Decision: Keeping this outliers for K-Means clustering would have been detrimental, as distance-based algorithms are highly sensitive to extreme values, risking the distortion of cluster centroids. Therefore, these players were removed to ensure stable, representative clusters for the "human" players.

The positive "Aliens" were not discarded. Instead, they were reintegrated in the final Moneyball analysis as a separate elite category using a prediction method based on the established cluster centroids. Furthermore, to validate our findings visually, we generated cluster maps both without outliers (to facilitate the interpretation of the core profiles) and with outliers included (to highlight the massive performance gap between the average player and the statistical elite).

METHODOLOGY

CLUSTER ANALYSIS

- **Hierarchical Clustering:**

Following the dimensionality reduction, we initiated the clustering process using Hierarchical Clustering. This exploratory step was critical to uncover the natural structure of the data without imposing a predefined number of groups. To determine the optimal number of clusters we employed a dual-criteria approach:

R-Squared Analysis: We examined the R-Squared plot to identify the point of diminishing returns where adding another cluster yields only a marginal increase in explained variance. As shown in Fig. 16, Ward's Method consistently yielded the highest R-Squared across all cluster solutions, validating its selection as the linkage method. The curve begins to plateau significantly after the fifth cluster, suggesting that a 5-cluster solution captures the majority of the variance efficiently.

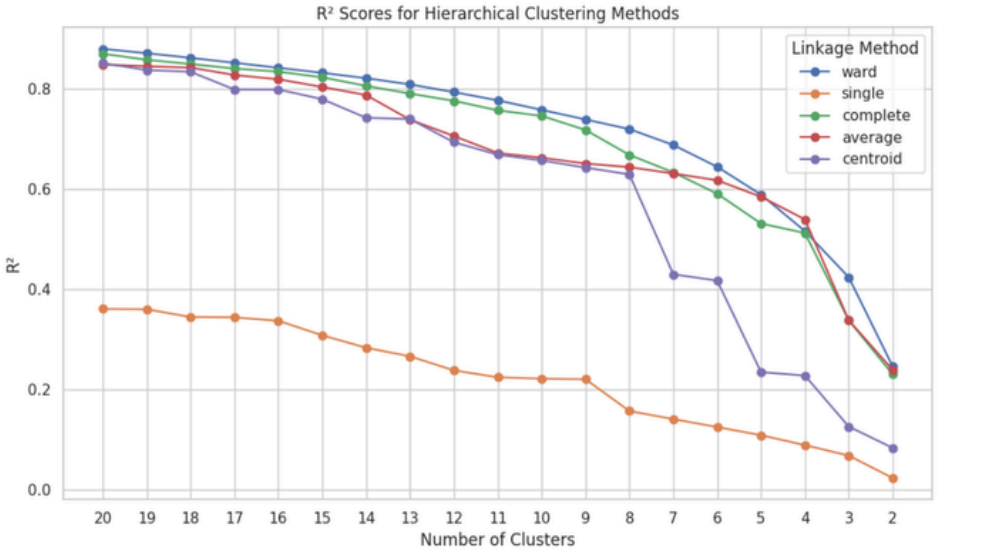


Fig. 17 - R-Squared scores

METHODOLOGY

CLUSTER ANALYSIS

Dendrogram Inspection: By analyzing the vertical distances in the dendrogram (Fig. 17), we looked for the largest "jump" or gap between merges. A cut-off threshold at a distance of approximately 6.5 naturally segments the data into five distinct groups. Merging beyond this point (e.g., to 4 clusters) would force disparate groups together, drastically increasing internal heterogeneity and loss of information.

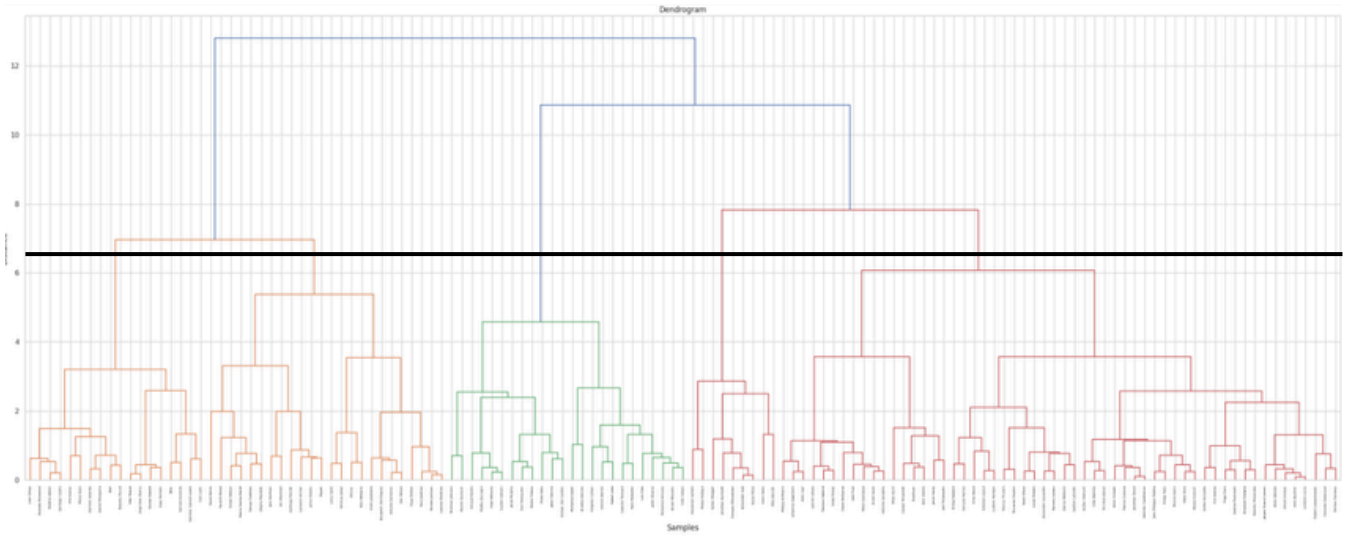


Fig. 18 - Dendrogram

METHODOLOGY

CLUSTER ANALYSIS

- **Non-Hierarchical Clustering:**

To optimize the homogeneity of our profiles, we proceeded to a Non-Hierarchical K-Means Analysis. Instead of random initialization, we used the centroids of the 5 clusters obtained from the hierarchical analysis as the initial seeds. This hybrid approach ensures the solution is both mathematically optimized and consistent with the natural structure identified earlier.

Model Validation: The final 5-cluster solution achieved an Overall R-Squared of 0.615, indicating that the clusters account for 61.5% of the total variance in player performance, a robust result for behavioral sports data.

Breaking down the variance explained by factor provides further insight into the cluster structure:

- **Playmaking & Workrate (R-Squared = 0.733):** This is the strongest discriminator. The high value indicates that the clusters are primarily defined by differences in creative output and defensive work rate.
- **Error Proneness (R-Squared = 0.621):** Also serves as a strong differentiator, effectively separating "safe" players from "high-risk" profiles.
- **Pure Finishing Threat (R-Squared= 0.491):** While still significant, this factor has the lowest explanatory power. This suggests that finishing ability is somewhat more distributed across different profiles (or concentrated in specific outliers) rather than being the sole definer of multiple groups.

METHODOLOGY

CLUSTER ANALYSIS

- **Big 5 Leagues - Cluster Titles:**

Based on the analysis of the cluster centroids (mean factor scores), we identified five distinct player profiles:

Cluster 0 - "Inefficient Possession Players" : Characterized by negative scores in both creative and finishing components, coupled with high Error Proneness. These players struggle to retain possession and offer limited offensive output, representing the least efficient profile in the dataset.

Cluster 1 - "High-Risk Creators" : Players with high Playmaking & Workrate but also high Error Proneness. These are energetic disruptors who drive the ball and press intensely but tend to be inconsistent in the final third, sacrificing possession security for creative volume.

Cluster 2 - "Elite Safe Playmakers" : The ideal creative profile: positive Playmaking combined with negative Error Proneness. These players generate high offensive volume while maintaining elite ball security, acting as the reliable engines of their teams.

Cluster 3 - "Complete Goalscorers" : The most exclusive cluster, containing players with Pure Finishing Threat > 1. This group encompasses both clinical low-volume finishers and complete forwards (e.g., Mbappé, Kane) who combine elite scoring output with significant offensive involvement.

METHODOLOGY

CLUSTER ANALYSIS

Cluster 4 - "Conservative / Low-Volume Forwards" : Players with near-zero finishing scores and low playmaking/error scores. This cluster represents a mix of efficient "poachers" (who touch the ball rarely but effectively) and average rotation players who adopt a low-risk, low-impact approach.

kmeans_cluster	Playmaking & Workrate	Pure Finishing Threat	Error Proneness
0	-0.632041	-0.441965	1.187215
1	0.910568	-0.777241	0.675451
2	1.136725	-0.157789	-0.863733
3	0.202041	1.752256	0.138066
4	-0.714118	0.092261	-0.498690

Fig. 19 - Cluster means for the big 5 leagues

METHODOLOGY

CLUSTER ANALYSIS

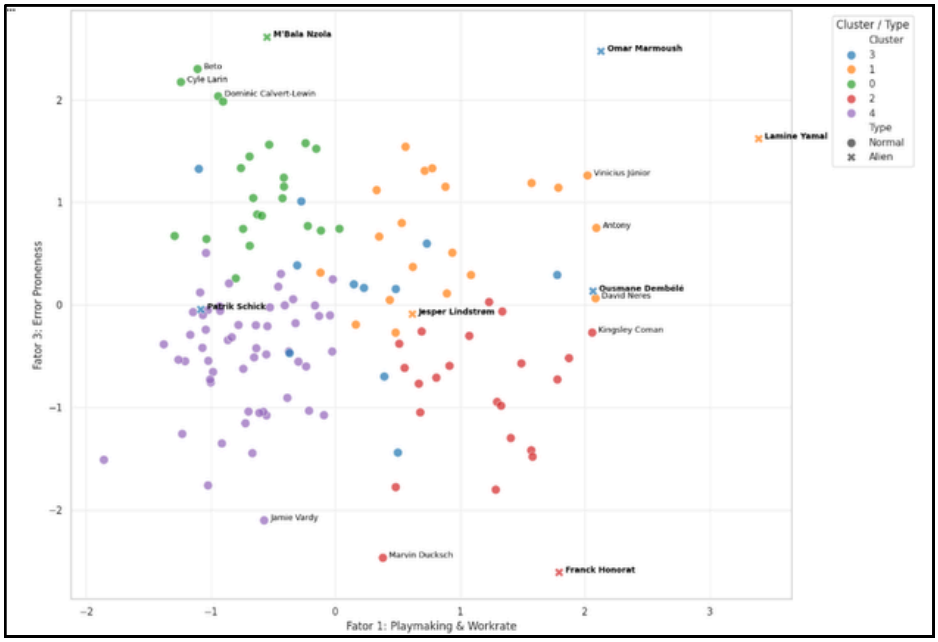


Fig. 20 - F1xF3 graph with outliers

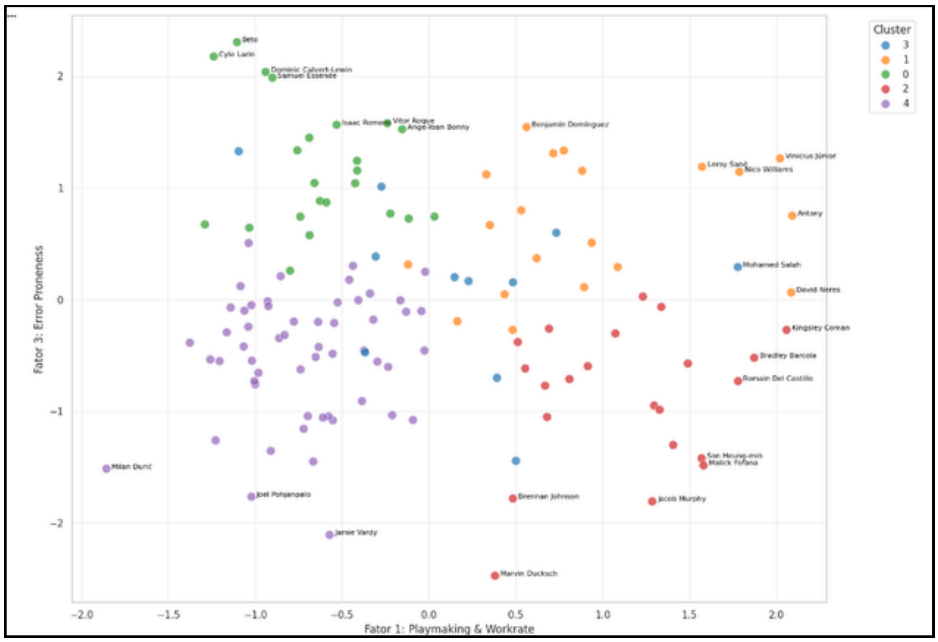


Fig. 21 - F1xF3 graph without outliers

METHODOLOGY

CLUSTER ANALYSIS

- **Secondary Leagues - Cluster Titles:**

The clustering of the secondary leagues revealed a slightly different structural configuration:

- **Cluster 0 - "Elite Safe Playmakers" :** The standout profile in these leagues. Extremely high Playmaking & Workrate combined with low errors. These are the central creators and high-volume contributors who dominate the offensive phase without losing possession.

- **Cluster 1 - "High-Risk Creators" :** Above-average playmaking and workrate paired with moderate error levels and low finishing. These are energetic attackers who contribute heavily to the game's flow but lack clinical efficiency or ball security compared to Cluster 0.

- **Cluster 2 - "Inefficient / Volatile Forwards" :** Marked by limited playmaking, average finishing, and high error rates. These are unpredictable attackers with volatile behavior in possession, often failing to translate risk into tangible output.

- **Cluster 3 - "Conservative / Low-Impact" :** Negative scores across all components. These players demonstrate limited involvement in build-up or finishing, exerting reduced offensive influence on the match.

- **Cluster 4 - "Clinical Poachers" :** Defined by low playmaking but high Finishing Threat and low errors. These are pure, efficient strikers focused almost exclusively on scoring rather than creation or build-up play.

METHODOLOGY

CLUSTER ANALYSIS

Cluster	Playmaking & Workrate	Pure Finishing Threat	Error Proneness
1	1.775903	-0.058271	-0.923493
2	0.928378	-0.366136	0.746827
3	-0.685376	-0.070285	0.651152
4	-0.484141	-1.005855	-0.828346
5	-0.503558	0.646991	-0.802674

Fig. 22 - Cluster means for the secondary leagues

METHODOLOGY

CLUSTER ANALYSIS

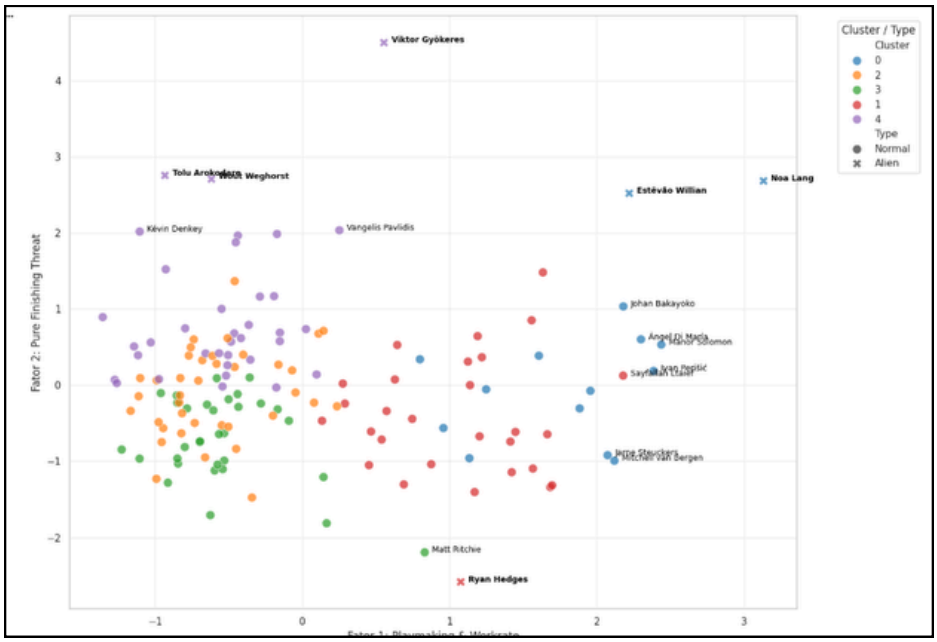


Fig. 23 - F1xF2 graph with outliers

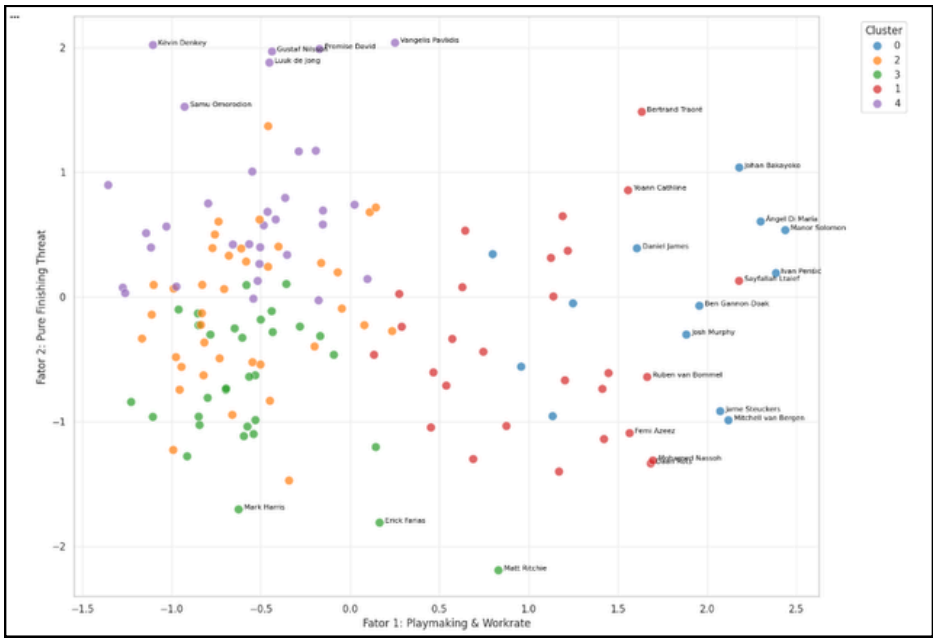


Fig. 24 - F1xF2 graph without outliers

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RESULTS

MONEYBALL ANALYSIS

To transition from statistical clustering to actionable **transfer targets**, we developed a robust selection framework designed to identify true market inefficiencies. Our **"Moneyball" algorithm** is built on three pillars:

Quality Criterion: The primary filter was performance, not price. We did not simply look for the cheapest players; we searched for Elite players who happen to be undervalued. A player was only considered for selection if their Factor Scores **exceeded the performance average of their specific cluster** (e.g., a "Safe Playmaker" must have a Playmaking Score > 1.0 and Error Proneness < -1.0).

Financial Criterion: To determine fair value, we opted for the Median (rather than the Mean) of each cluster's **market value**. This decision **mitigates** the impact of extreme outliers (like Haaland or Mbappé) who artificially inflate the average price. By using the median, we established a fairer baseline for the "normal price" of elite performance within each profile.

Age Context: The final analysis integrates the age dimension, segmenting opportunities into 'Development' (age 23), 'Prime' (24-29), and 'Veteran' (age 30). This allows us to tailor recommendations to the specific investment profile of a potential buyer, distinguishing between long-term assets and immediate short-term fixes.

RESULTS

MONEYBALL ANALYSIS

- **Target Scope:**

Major leagues - Our analysis focused exclusively on Clusters 1, 2, 3, and 4. Cluster 0 ("Inefficient Players") was removed from the Moneyball analysis as these players do not meet the minimum performance standards required for elite investment.

Players below represent proven elite quality currently undervalued by the market:

Secondary leagues - We narrowed our focus to Clusters 0, 3, and 4. Excluding erratic or inconsistent profiles (clusters 1 and 2) to maximize the reliability of the recommended transfer targets.

- **Cluster-Specific Thresholds (The "Elite" Filter):**

Instead of a generic filter, we applied logic adapted to the tactical role of each cluster (derived from our SAS algorithm):

- **For High-Risk Creators (Clusters 1) :** Must have a Playmaking & Workrate score > 1.5 to justify the higher error rate.
- **For Elite/Safe Playmakers (Cluster 2 of the big5 and Cluster 0 of the secondary leagues):** Must maintain a Playmaking score > 1.0 while keeping Error Proneness < -1.0.
- **For Goalscorers/Poachers (Clusters 4):** Must have a Pure Finishing Threat score > 1.0.
- **For Complete Forwards (Cluster 3 of the Big5):** All players in this cluster were considered elite due to their balanced high scores.

RESULTS

MONEYBALL ANALYSIS

Player	Cluster	Age	Market Value	Median M.Value	Playmaking & Workrate	Pure Finishing Threat	Error Proneness
David Neres	1	28	30,000,000	35,000,000	2.0857185197023607	-1.604434151598512	0.06396320858802776
Leroy Sané	1	29	32,000,000	35,000,000	1.5724950854847624	0.5568759939410781	1.188918034290421
Antony	1	25	35,000,000	35,000,000	2.0921460445691187	-0.4653140117729963	0.7504694872278506
Franck Honorat	2*	28	12,000,000	25,000,000	1.793702724594772	-0.8515566943733962	-2.6117712446133097
Jacob Murphy	2	29	16,000,000	25,000,000	1.2855847366416606	-0.26406720268970413	-1.8044357056940274
Son Heung-min	2	33	20,000,000	25,000,000	1.570073152697007	-0.13754894607554025	-1.4198266738816354
Patrik Schick	3*	29	27,000,000	45,000,000	-1.0793328011075667	2.7969595455038117	-0.04495589548276621
Ayoze Pérez	3	34	10,000,000	45,000,000	0.14772485603226512	1.4385604845577387	0.20047126682054975
Georges Mikautadze	3	23	22,000,000	45,000,000	0.4841614987770671	1.4140315824665517	0.1544895655821611
Alexander Sørloth	3	28	25,000,000	45,000,000	-1.0950588598234714	2.3008369764419854	1.326915625781387
Mika Biereth	3	22	30,000,000	45,000,000	-0.3676055391531001	2.139933259294185	-0.4715611686890044
Jonathan Burkardt	3	22	35,000,000	45,000,000	-0.3063228836746306	1.281053029525541	0.38541166567970936
Mateo Retegui	3	26	45,000,000	45,000,000	-0.2735038043537537	2.28734582780996	1.0103242575411544
Chris Wood	4	37	10,000,000	31,000,000	-1.2282626915261177	1.0752144928368346	-1.259679146477993
Robert Lewandowski	4	37	12,000,000	31,000,000	-1.0845832701801355	1.1339521915940418	0.1206051688317482
Ermedin Demirović	4	27	22,000,000	31,000,000	-0.55429046170497	1.1930806429080119	-0.4842145712191405

Fig. 25 - Resulting table from the Big5 leagues

Cluster (*): Indicates that the cluster assignment was predicted by the model. These players were identified as extreme statistical outliers ("Aliens") and were excluded from the initial training of the K-Means algorithm to prevent centroid distortion.

Status/Age Symbols:

- 👽 **Alien:** A multidimensional outlier exhibiting absolute elite statistical performance (quality benchmark).
- 💎 **Gem (Development):** Players aged 23 or younger (High potential for future valuation growth).
- 🥈 **Silver (Prime):** Players aged between 24 and 29 (Peak performance phase).
- 🥉 **Bronze (Veteran):** Players aged 30 or older (Short-term investment/Experience).

RESULTS

MONEYBALL ANALYSIS

The top Picks from Big 5 Leagues

1. Franck Honorat - Profile: **Elite-Safe Playmaker**
Age: 28 (Silver - Prime) ; "Alien" Status
Market Value: €12M (vs. Cluster Median: €25M)
Our overview:
Honorat is arguably the most efficient winger in our dataset. With a Playmaking Score of 2.04 and an incredibly low Error Proneness of -2.93, he offers elite creativity with virtually zero waste. Having established himself in the Bundesliga after a strong stint at Brest, he creates danger without losing possession. At €12M less than half the median price for players of this caliber he is a low-risk, high-reward acquisition for any team needing immediate stability and creativity.

2. Patrik Schick – Profile: **Prolific Goalscorer**
Age: 29 (Prime/Late Prime) ; "Alien" Status
Market Value: €27M (vs. Cluster Median: €45M)
Our overview:
A classic "Alien" in our model. Schick does not participate in the build-up (Playmaking -1.31), but his Finishing Threat (2.68) is world-class. His market value has likely dipped due to injury history, creating a distinct market inefficiency. For a club that needs a pure "fox in the box" to convert chances immediately, paying €27M for a striker who statistically performs closer to the €60M+ bracket is a strategic masterstroke.

RESULTS

MONEYBALL ANALYSIS

3. Mika Biereth - Profile**: The Young Finishing Menace**
Age: 22 (Gold – Young)
Market Value: €30M (vs. Cluster Median: €45M)
Our overview:
Biereth is one of the clearest undervalued prospects in the entire dataset. With a Pure Finishing Threat score of 2.14, he ranks among the top performers in his cluster despite being significantly younger than most peers. His price is shockingly low, €15M below the cluster median, which makes him a high-upside, low-risk investment. Clubs searching for a young striker with elite scoring instincts and room for rapid development should circle his name in bold.

4. Georges Mikautadze - Profile: **The Creative Dual Threat**
Age: 23 (Gold – Young)
Market Value: €22M (vs. Cluster Median: €45M)
Our overview:
Mikautadze combines playmaking contribution (0.48 in Playmaking & Workrate) with strong scoring instincts (1.14 in Finishing), giving him one of the most balanced and valuable profiles among modern forwards. His ability to create danger between lines while maintaining scoring efficiency is extremely rare for a player priced €23M below his cluster's median. He is the definition of a bargain: a dual-threat attacker whose market value simply has not caught up with his output.

RESULTS

MONEYBALL ANALYSIS

5. Antony - Profile: **The High-Output Creative Winger**
Age: 25 (Silver – Prime)
Market Value: €35M (vs. Cluster Median: €35M)
Our overview:
Contrary to popular belief, Antony's Playmaking & Workrate score of 2.09 places him among the highest-output creative wingers in the cluster. His influence in ball progression, chance creation, and high-intensity actions is elite, and his price now aligns perfectly with the cluster median. That means he is no longer a **"discount pick"**, instead, he becomes a fair value option: you pay the standard market rate, and you receive a winger with premium creation metrics and strong overall involvement. Teams seeking a **high-activity wide creator** with reliable buildup quality will find Antony an analytically sound, risk-neutral acquisition.

RESULTS

MONEYBALL ANALYSIS

Resulting Table for the Secondary 5 Leagues:

Player	Cluster	Age	Market Value	Median M.Value	Playmaking & Workrate	Pure Finishing Threat	Error Proneness
Ivan Perišić	0	🥉 36	1,500,000	3,000,000	2.3859182195632833	0.19016031467589414	-2.8770378142947726
Mitchell van Bergen	0	🥈 26	1,500,000	3,000,000	2.1195074046411055	-0.990767637392752	-1.310325984962727
Telmo Arcanjo	0	🥈 24	3,000,000	3,000,000	1.1350366188107188	-0.9578662905770601	-1.1933232135208869
Mohamed Nassoh	1	💎 23	700,000	4,000,000	1.6964697392223431	-1.3137548304544113	0.3342374910157308
Femi Azeez	1	🥈 24	1,200,000	4,000,000	1.5662078879414667	-1.094052181951639	1.3192541583056971
Sayfallah Ltaief	1	🥈 25	1,800,000	4,000,000	2.179158138832122	0.1277633911506941	1.980132661977541
Daan Rots	1	💎 23	4,000,000	4,000,000	1.684599759504361	-1.3376485941531322	0.08059392024487304
Wout Weghorst	4*	👽 32	4,000,000	11,000,000	-0.6167043393713121	2.7114144802428193	-1.550718611133314
Luuk de Jong	4	🥉 35	2,500,000	11,000,000	-0.45150185956564554	1.8792380357977596	0.1587579728554291
Gustaf Nilsson	4	🥈 28	5,000,000	11,000,000	-0.4371606970002328	1.9691297184410614	0.5274694327306383
Haji Wright	4	🥈 27	8,000,000	11,000,000	-0.28721950972073734	1.1666276157461144	-0.08595254971793036
Tyrese Campbell	4	🥈 25	8,000,000	11,000,000	-0.5476321651244634	1.0048940101454915	-1.3761758596318718
Promise David	4	🥈 24	11,000,000	11,000,000	-0.17135632002026	1.989101019002945	-0.23602144600248343

Fig. 26 - Resulting table from the secondary leagues

Cluster (*): Indicates that the cluster assignment was predicted by the model. These players were identified as extreme statistical outliers ("Aliens") and were excluded from the initial training of the K-Means algorithm to prevent centroid distortion.

Status/Age Symbols:

- 👽 **Alien:** A multidimensional outlier exhibiting absolute elite statistical performance (quality benchmark).
- 💎 **Gem (Development):** Players aged 23 or younger (High potential for future valuation growth).
- 🥈 **Silver (Prime):** Players aged between 24 and 29 (Peak performance phase).
- 🥉 **Bronze (Veteran):** Players aged 30 or older (Short-term investment/Experience).

RESULTS

MONEYBALL ANALYSIS

The Top 5 Picks from Secondary Leagues

1. Mohamed Nassoh - Profile: **The Hidden Gem**
Age: 23 (Gold – Young)
Market Value: €700k (vs. Cluster Median: €4M)
Our overview:
Nassoh is the **single best value-for-money** midfielder/forward hybrid in the entire secondary-league sample. His Playmaking & Workrate score of 1.69 puts him among the **elite creators** of the dataset, while his cost sits at barely 1/6th of his cluster's median price. He's the rare pick who is both analytically elite and ridiculously cheap.

2. Mitchell van Bergen - Profile: **The Undervalued Creator**
Age: 26 (Silver – Prime)
Market Value: €1.5M (vs. Cluster Median: €3M)
Our Overview:
Van Bergen posts the second-highest Playmaking & Workrate score in the dataset (2.12) with one of the lowest Error Proneness scores aswell (-1.31), which is absurd considering his price is half of the cluster median. He's the classic overperforming winger who doesn't get recognition because he plays in the Eredivisie mid-table, but the data sees him clearly. For teams needing a high-motor wide player who produces volume actions and links play, he's **one of the biggest bargains available.**

RESULTS

MONEYBALL ANALYSIS

3. Wout Weghorst - Profile: **Clinical Poacher**
Age: 32 (Veteran) ; “Alien Status”
Market Value: €3.5M (vs. Cluster Median: €11M)
Our overview:
An "Alien" in the data. Weghorst creates nothing (-0.61), but his Finishing (2.71) and ball retention (Error -1.55) are off the charts. At 32, he offers no resale value, but for €3.5M, he guarantees elite finishing for 1-2 seasons. The **perfect short-term solution** for a team needing instant goals.

4. Promise David - Profile: **Clinical Poacher**
Age: 24 (Prime)
Market Value: €11M (vs. Cluster Median: €11M)
Our overview:
A solid investment. At 24, he is entering his prime with a strong Finishing Score (1.98). While not a "steal" at €11M, his age and consistent finishing metrics suggest he can hold his value or appreciate if he moves to a bigger league. A safe, logical transfer target.

RESULTS

MONEYBALL ANALYSIS

5. Sayfallah Ltaief - Profile: **The Creative Wildcard**
Age: 25 (Prime)
Market Value: €1.8M (vs. Cluster Median: €4M)
Our Overview:
Ltaief combines strong creation metrics (2.17 in Playmaking & Workrate) with moderate Finishing Threat and higher risk in possession, but for under €2M, this is exactly the type of upside swing you want. He is an on-ball difference maker, capable of destabilizing low blocks and progressing the ball through dribbles. For clubs searching for a low-cost creator who bends defensive structures, even though he has the highest Error Proneness Score, Ltaief is a perfect analytics-based discovery.

DISCUSSION

OVERVIEW

The results of our analysis reveal clear and interpretable patterns in forward profiles across both major and secondary leagues. By applying Principal Component Factoring (PCF) with Varimax rotation, we successfully compressed a large set of performance indicators into three meaningful components, Playmaking & Workrate, Pure Finishing Threat, and Error Proneness, which captured the essential behavioral differences between attacking players. The rotated structure allowed us to clearly distinguish wingers, who typically display high involvement and creation, from strikers, who cluster around direct finishing actions.

Crucially, the identification and removal of outliers prior to clustering provided a vital stabilizing effect. Extreme players such as Ousmane Dembélé, Lamine Yamal, or Viktor Gyökeres exerted disproportionate influence on the distance-based clustering methods, which would have distorted cluster boundaries. By temporarily removing these “Aliens,” we ensured more homogeneous clusters for the general population. However, these outliers were not discarded, they were reintegrated into the final analysis as elite benchmarks. The detection of these specific players serves as a powerful validation of our model's accuracy: by statistically isolating the 2024/2025 Ballon d'Or winner (Dembélé), the runner-up (Yamal), and the European Golden Shoe runner-up (Gyökeres) as anomalies, the algorithm successfully proved its ability to distinguish absolute world-class performance from standard professional output without human bias.

DISCUSSION

OVERVIEW

Clustering results in both datasets revealed five coherent player typologies per league group. For the Big 5 leagues, clusters ranged from high-volume creators to pure penalty-box finishers and low-risk technical wingers. Similar archetypes emerged in secondary leagues, suggesting that forward roles remain consistent across competition levels even if performance magnitudes differ. These clusters allowed us to identify market inefficiencies, particularly players whose performance metrics were significantly above their cluster's median while their price was below it.

DISCUSSION

SPECIAL CASES

Beyond identifying market targets, the detailed analysis of outliers and the comparison between the two ecosystems (Big 5 vs. Secondary Leagues) revealed fundamental structural patterns regarding talent in European football.

- **The "Marmoush Paradox" : Statistics vs. Context**

Omar Marmoush represents the most complex and educational case study in our dataset, illustrating the tension between pure talent identification and contextual adaptation.

The Multidimensional Outlier: Statistically, Marmoush (based on his Eintracht Frankfurt data) was the most complete player in the sample. He presented elite scores in both Playmaking and Finishing Threat, combined with extremely high Error Proneness. He is the definition of "Productive Chaos": a player who attempts everything, fails often, but generates unstoppable offensive volume.

The Manchester City Lesson: His January transfer to Manchester City (reflected in the drop in minutes that excluded him from the second half of the sample) serves as a warning. Although our model validated the quality that motivated the purchase (elite scores), his subsequent performance in England, solid, but not dominant, reminds us that football is not played on a spreadsheet. The statistical model excels at identifying performance potential (what he did in Germany), but it cannot predict the adaptation friction to a more intense league or a more rigid tactical system (such as Guardiola's), where his high "Error Proneness" may be less tolerated.

DISCUSSION

SPECIAL CASES

- **Market Validation: The "Aliens" of the Secondary Leagues**

The robustness of our predictive model is strongly validated by the actual market movement of players identified as "Aliens" in the Secondary Leagues.

The Statistical "Prophecy": The vast majority of identified positive outliers (Viktor Gyökeres, Estêvão Willian, Noa Lang, Tolu Arokodare) starred in high-profile transfers and saw their market value explode to "Big 5" levels in the summer of 24/25. This confirms that our PCF/Clustering algorithm aligns with world-class elite scouting.

The Weghorst Case: The exception is Wout Weghorst. Being a veteran, he did not realize a "galactic" transfer, but the model correctly identified him as a specific utility profile (similar to Luuk de Jong's role at Barcelona/PSV/Porto). He proves that the model detects immediate quality, regardless of future resale potential.

DISCUSSION

SPECIAL CASES

- **Structural Divergence: Depth of Elite Talent**

When comparing the cluster compositions between the two ecosystems, a clear structural difference emerges regarding the depth of elite talent:

In Big 5 Leagues: We identified Cluster 3 ("Complete Goalscorers"), a robust group densely populated by finishing specialists operating at an elite statistical level (e.g., Kane, Mbappé). This demonstrates that the Big 5 leagues possess a deep pool of high-end talent; there are enough elite finishers to form an entire statistical cluster on their own.

In Secondary Leagues: This specific "Elite Cluster" is virtually non-existent among the normal population. Once the statistical outliers (the "Aliens" like Gyökeres) are removed, the remaining "best" cluster consists of good, functional strikers, but they lack the overwhelming statistical output seen in the Big 5's top cluster. Therefore, the talent distribution in Secondary Leagues is "top-heavy" dominated by a tiny handful of "Aliens" who perform far above the rest, whereas the Big 5 leagues maintain a broad, consistent layer of elite quality that extends beyond just the top few names.

DISCUSSION

LIMITATIONS

While the proposed framework offers a robust quantitative method for identifying market inefficiencies, it is essential to acknowledge the limitations inherent to a purely statistical approach:

Tactical Blindness (Output vs. Role): The model evaluates players based on output metrics (what they did) without context for tactical instructions (what they were asked to do). For instance, a striker with low "Playmaking" scores may possess the technical ability to link play but be tactically instructed to remain central and pin opposition defenders. The algorithm interprets this as low involvement or limited skill, whereas a scout would correctly identify it as tactical discipline.

The "Intangibles" Gap: Our Factor Analysis captures measurable on-ball actions and defensive work rate but remains blind to critical intangible qualities. Attributes such as off-ball movement, leadership, decision-making under pressure, and adaptability to new environments are vital determinants of transfer success but cannot be fully quantified by event data alone.

Sample Size Constraints: The rigorous filter of >35% minutes played, while necessary to ensure statistical stability and reduce variance, inevitably excluded specific player profiles. Potential "super-subs" or high-impact January signings (such as Omar Marmoush's stint at Manchester City) were omitted due to insufficient volume, potentially causing the model to miss niche market opportunities that did not meet the volume threshold.

DISCUSSION

LIMITATIONS

Injury Exclusion Bias: Still on the strict minutes threshold applied (>35% of the season), it prioritizes statistical stability but creates a blind spot regarding injured talent. Elite players who suffered long-term injuries during the 2024/2025 season were automatically excluded from the dataset. Consequently, the model fails to identify high-quality players whose market value may have dropped due to temporary unavailability, representing potential high-reward opportunities for medical departments willing to manage the risk.

Multidimensionality of Market Value: Our valuation model simplifies the concept of "Fair Price" primarily to a function of On-Field Performance and Age. However, real-world market value is driven by a broader set of variables not captured in this study, such as contract duration (players with <1 year left are artificially cheaper), media profile/commercial appeal, nationality premiums (e.g., the "Homegrown" tax), and the selling club's financial health. Therefore, a player identified as a "Hidden Bargain" might be cheap due to an expiring contract rather than a pure market inefficiency regarding their talent.

Positional Classification Bias: Our dataset construction relied strictly on the "Forward" (FW) classification provided by FBref. However, positional taxonomy in modern football is increasingly fluid. Many high-output offensive players, such as Raphinha, Florian Wirtz, or Cole Palmer, operate effectively as forwards but are often categorized as "Midfielders" (MF) or "Attacking Midfielders" within database taxonomies. Consequently, our analysis excludes these hybrid profiles, potentially missing elite offensive contributors who deliver "forward-like" statistical output but fall outside the rigid positional filter used for data collection.

CONCLUSION

This project successfully combined PCF, outlier diagnostics, and clustering methods to build a data-driven framework for evaluating forwards across European football leagues. The three latent components uncovered, Playmaking & Workrate, Pure Finishing Threat, and Error Proneness, proved effective in differentiating between forward archetypes and highlighting the strengths and weaknesses that define a player's on-pitch behavior.

The clustering process produced intuitive player groups and allowed us to identify five standout value picks in each dataset, highlighting players whose performance levels exceed what would be expected for their price category. These insights demonstrate how statistical modeling can reveal hidden opportunities in the transfer market, especially when applied to leagues where scouting resources are traditionally limited.

From a practical standpoint, the results provide clubs with:

- A **structured way to categorize forwards** based on underlying playing style;
- A method to **detect undervalued talent** relative to their peer group;
- An evidence-supported approach to **risk reduction** when evaluating potential signings;
- A foundation for integrating analytics with traditional scouting practices.

In summary, the analysis shows that even with varied data sources and inherent limitations, advanced statistical methods can meaningfully contribute to recruitment decisions. By quantifying player behavior and comparing performance within stylistically coherent groups, clubs can identify market inefficiencies, target players who offer both quality and affordability, and ultimately make more informed strategic investments.

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APPENDIX

Major leagues (Big5):

Dataset: Big5

Dataset: MoneyBall Big5

Statistical Analysis Python Code of the Big5 Leagues

Big5 MoneyBall SAS Code

Big5 Merge SAS Code

Secondary Leagues (Low5):

Dataset: Low5

Dataset: MoneyBall Low5

Statistical Analysis Python Code of the Low5 Leagues

Low5 MoneyBall SAS Code

Low5 Merge SAS Code

All Leagues (Big5 & Low5):

Different tables of FBref - Merge SAS Code